Neutral density evaluation from wave damping

The ion acoustic wave amplitude decays exponentially,

$$A(x) = A_0 e^{-x/L},$$

and the measured damping length is

$$L = 0.05 \text{ m}.$$

Since the dominant damping mechanism in weakly ionized argon gas is ion–neutral momentum transfer, the damping length is related to the ion–neutral collision frequency by

$$\nu_{in} = \frac{V_p}{L}.$$

Using the measured ion acoustic phase velocity

$$V_p = 375.7 \text{ m/s},$$

we obtain

$$\nu_{in} = \frac{375.7}{0.05} = 7.51 \times 10^3 \text{ s}^{-1}.$$

The ion–neutral collision frequency is also given by

$$\nu_{in} = n_n \sigma v_{th},$$

where σ is the ion–neutral collision cross-section and v_{th} is the neutral thermal velocity. For low-energy $\mathrm{Ar}^+ - \mathrm{Ar}$ collisions, we use

$$\sigma = 1 \times 10^{-18} \text{ m}^2,$$

consistent with momentum-transfer data for argon in low-temperature plasma databases (e.g., LXCat).

The neutral gas temperature is taken as

$$T = 300 \text{ K},$$

because in weakly ionized plasmas the neutral population remains near room temperature: electron-neutral energy transfer is negligible at low ionization degree.

The neutral thermal velocity is therefore

$$v_{th} = \sqrt{\frac{8k_B T}{\pi m_{\mathrm{Ar}}}} = \sqrt{\frac{8(1.380649 \times 10^{-23})(300)}{\pi\,(6.6335 \times 10^{-26})}} = 3.99 \times 10^2 \text{ m/s}.$$

Thus, the neutral density obtained from the measured damping is

$$n_n = \frac{\nu_{in}}{\sigma v_{th}} = \frac{7.51 \times 10^3}{(1 \times 10^{-18})(3.99 \times 10^2)} = 1.88 \times 10^{19} \text{ m}^{-3}.$$

Ideal gas comparison

The chamber pressure during the experiment was

$$p = 8.1 \times 10^{-4} \text{ mbar} = 8.1 \times 10^{-4} \times 100 \text{ Pa} = 0.081 \text{ Pa}.$$

The ideal-gas neutral density is therefore

$$n_{ideal} = \frac{p}{k_B T} = \frac{0.081}{(1.380649 \times 10^{-23})(300)} = 1.96 \times 10^{19} \text{ m}^{-3}.$$

Final result

$$n_{n,damping} \approx 1.9 \times 10^{19} \text{ m}^{-3} \qquad n_{n,ideal} \approx 2.0 \times 10^{19} \text{ m}^{-3}$$